

Azure Cost Management with Budget & Anomaly Alerting

PROJECT REQUIREMENTS DOCUMENT

24CC3046-P016 | Cloud Computing

Azure Resources



Cost Data / Usage



Cost Management + Budget + Anomaly Detection



Email Notifications → Team Investigation & Action

Budget monitoring • Cost analysis • Ownership tagging • Anomaly alerting

1. Introduction

This document defines the functional, technical, infrastructure, monitoring, and validation requirements for the project “Azure Cost Management with Budget & Anomaly Alerting.” The solution uses Azure Cost Management to monitor subscription spending, apply a monthly budget, notify responsible owners at configured thresholds, identify unusual cost behaviour, and provide a structured view for monthly cost analysis.

2. Problem Definition

Cloud resources can generate cost without clear ownership or timely visibility. A centralized cost-monitoring approach is required to track spend against an approved budget, alert owners when thresholds are reached, review service-level costs, and detect unusual spending behaviour. Untagged resources can also make ownership and chargeback analysis incomplete.

3. Project Objectives

- Create a small Azure environment suitable for an academic demonstration using an Azure for Students subscription.
- Organize project resources in a dedicated resource group.
- Apply Department, Project, and Owner tags for ownership and accountability.
- Create a monthly subscription budget and configure actual-cost threshold alerts.
- Use Cost Analysis to review monthly spending by service and investigate cost drivers when cost data is available.
- Configure an anomaly alert rule that notifies project owners when Azure detects unusual cost behaviour.
- Collect Azure Portal screenshots as implementation evidence.

4. Functional Requirements

ID	Requirement	Description
FR-01	Azure environment	The system shall use an eligible Azure subscription to host the project resources.
FR-02	Resource organization	The system shall use a dedicated resource group for the project resources.
FR-03	Ownership tagging	Resources shall use Department, Project, and Owner tags for accountability.
FR-04	Budget	The system shall provide a monthly subscription budget for spending control and visibility.
FR-05	Threshold alerts	The system shall notify configured recipients when actual cost reaches the defined budget thresholds.
FR-06	Cost analysis	The system shall provide a Cost Analysis view grouped by service with monthly granularity.
FR-07	Anomaly detection	The system shall use an anomaly alert rule to identify unusual cost behaviour.
FR-08	Notification	The system shall send anomaly notifications to configured project owners.
FR-09	Monthly drivers	The system shall support identification of the top five monthly cost drivers when cost data is available.
FR-10	Evidence	The project shall collect screenshots of the implemented Azure configuration for validation.

5. Infrastructure Requirements

Component	Status	Requirement
Azure subscription	Required	Eligible Azure for Students/Free subscription within available credits and limits.
Resource Group	Required	Dedicated resource group for project resources.
Storage Account	Required	Small Azure StorageV2 resource for demonstration and cost monitoring.
Tags	Required	Department, Project, and Owner tags applied to project resources.
Cost Management	Required	Cost Analysis, Budgets, and Cost Alerts capabilities.
Monthly Budget	Required	₹10 monthly budget for the current academic demonstration.
Email recipients	Required	Project-owner email addresses configured for alerts.
Internet connection	Required	Existing Internet connection for Azure Portal access.

6. High-Level Architecture

The high-level solution connects Azure resources and their usage to Azure Cost Management. Cost Analysis provides visibility, the monthly budget evaluates actual spend against thresholds, and anomaly detection identifies unusual cost behaviour. Email notifications inform the responsible owners, who review the data and take corrective or investigative action.

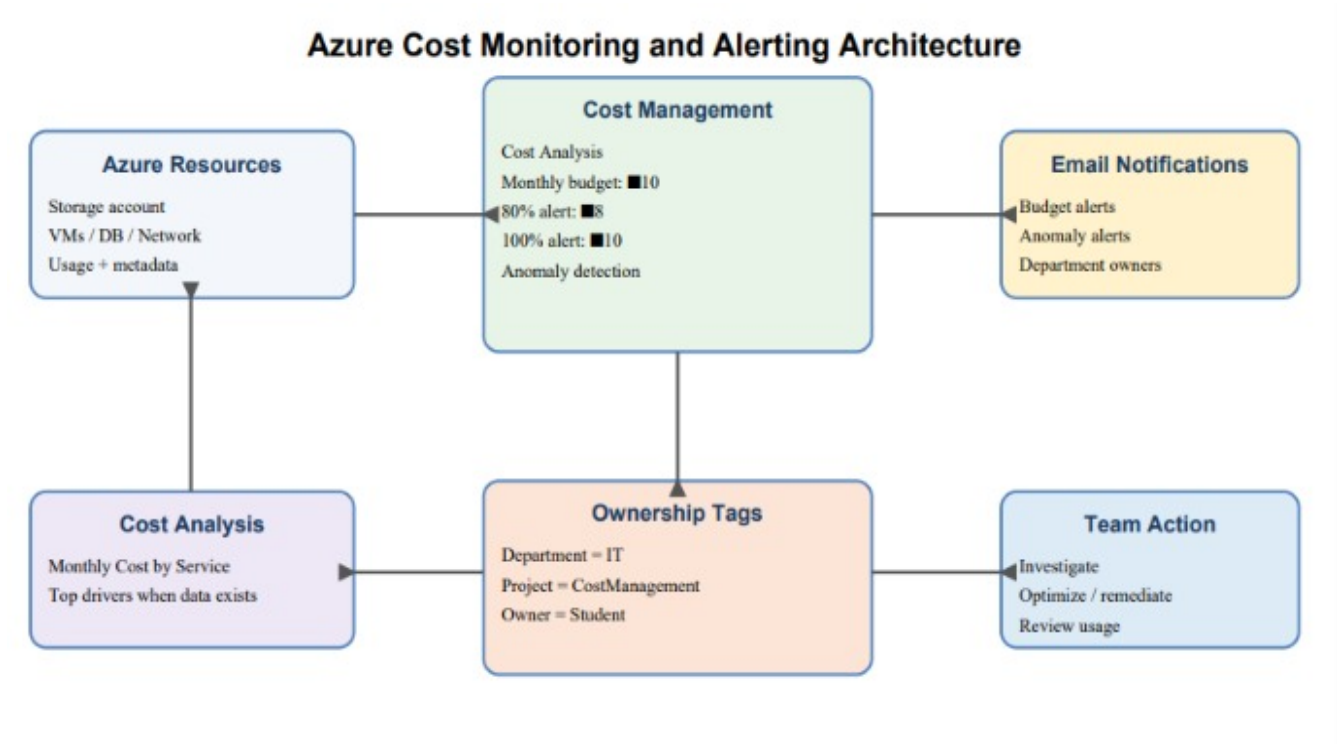


Figure 1. High-level architecture for the proposed Azure Cost Management and alerting solution.

7. Implementation Requirements

High-level implementation sequence

Step	Action	Implementation
1	Create Azure	Use the Azure for Students subscription.
2	Create Resource Group	Create RG-CostManagement for the project.
3	Create Resource	Deploy costalertstorage001 in East Asia.
4	Apply Tags	Department = IT; Project = CostManagement; Owner = Student.
5	Create Budget	Create MonthlyCostBudget with a ₹10 monthly limit.
6	Configure Alerts	Set actual-cost thresholds at 80% (₹8) and 100% (₹10).
7	Configure Anomaly Rule	Activate Daily anomaly by resource group and email recipients.
8	Analyze Costs	Use Monthly Cost by Service for monthly service-level review.

Implementation validation shall include:

- Verify the resource group contains the project resource.
- Verify the storage account shows the required ownership tags.
- Verify the monthly budget shows ₹10 and the 80%/100% actual-cost thresholds.
- Verify the Cost Analysis view is configured for monthly service-level analysis.
- Verify the anomaly alert rule is shown as Active with the expected condition and recipients.
- Collect screenshots for the academic project record.

8. Non-Functional Requirements

Quality Attribute	Requirement
Availability	The Azure monitoring configuration should remain accessible while the required subscription and services are available.
Scalability	The design should allow additional Azure resources and services to be included in cost analysis.
Reliability	Budget thresholds, anomaly rules, and recipient configuration should be checked after setup.
Security	Azure access and project resources should use appropriate permissions and secure configuration.
Performance	Cost Analysis should provide a usable view for monthly investigation when billing data is available.
Manageability	Resources, tags, budgets, views, and alert rules should be organized so students can monitor and maintain them.
Recoverability	Configuration details and screenshots should be documented so the academic setup can be recreated.

9. Use-Case Requirements

Budget Monitoring: Actual subscription cost is compared with the configured monthly budget.

Threshold Alerting: Email notifications are configured for 80% (₹8) and 100% (₹10) of the monthly budget.

Anomaly Detection: Azure Cost Management evaluates the configured anomaly rule and can notify the project owners when unusual cost behaviour is detected.

Cost Analysis: The Monthly Cost by Service view supports service-level monthly review.

Ownership: Tags identify Department, Project, and Owner information for project resources.

Investigation: When an alert occurs, the team reviews Cost Analysis and the related resource/service before taking action.

10. Subscription Requirement and Alternatives

The project distinguishes between the Azure subscription used to create resources and the payment model for those resources. The current implementation uses Azure for Students and keeps the demonstration small. If a required service or usage level is outside the available student limits, a limited demonstration or an institutional/Pay-As-You-Go subscription may be considered.

Service / Resource	If Student Limits Are Enough	Alternative if Paid Usage Is Required
Azure Subscription	Use Azure for Students within available credit/limits.	Use institutional or Pay-As-You-Go subscription.
Storage Account	Use a small StorageV2 resource and limit unnecessary data storage.	Use paid storage configuration for larger or longer-running data.
Cost Management	Use Cost Analysis, Budgets, and Cost Alerts within the subscription.	Continue using the monitoring features under the applicable paid subscription.
Anomaly Alerting	Use the configured anomaly rule and email notification.	Continue anomaly monitoring for larger production subscriptions.
Monitoring	Use only required monitoring views and evidence collection.	Use expanded monitoring/logging if operational needs increase.

11. Team Members

Student ID	Team Member
2400031197	T. Naga Bala Nandini
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2400031152	K. Eswar Naga Sai

12. Project Scope

This project is designed as a small-scale academic demonstration using an Azure for Students subscription. The core scope includes a dedicated resource group, tagged storage resource, subscription-level monthly budget, 80% and 100% actual-cost alerts, Cost Analysis, anomaly alerting, email notification, and documented validation evidence.

13. Azure Services / Resources and Usage

Service / Resource	Usage in This Project	PAYG / Usage Consideration	Benefit to Project
Azure Cost Management	Provides Cost Analysis, Budgets, and Cost Alerts	Billing/cost-management features depend on the Azure subscription and billing period.	Centralize the Azure visibility and control.
Budgets	MonthlyCostBudget with ₹10 limit and actual-cost alerts	Necessary for resource deployment; Early cost alerts at 80% and 100% threshold at ₹10.	Early cost alerts at 80% and 100% threshold at ₹10.
Cost Analysis	Monthly Cost by Service view for September 2026	At 2026 billing data appears as Azure supports cost	Supports cost service/cost-driver investigation.
Cost Alerts / Anomaly	Daily anomaly by resource group rule; Active and email notification	Alerts depend on available data; Detects abnormal spending behaviour.	Detects abnormal spending behaviour.
Storage Account	costalertstorage001; StorageV2; Standard; \$0.065 per GB per month	Safe storage/operations may incur charges	Provides a storage resource for demonstration and testing.
Tags	Department=IT; Project=CostManagement; No resource charge for tags	Improves ownership and allocation analysis.	Improves ownership and allocation analysis.

14. Cost Monitoring and Validation

- The resource group contains the project resource and can be shown in the Azure Portal.
- The storage account displays the required Department, Project, and Owner tags.
- The MonthlyCostBudget displays a ₹10 amount and actual-cost alerts at 80% and 100%.
- The Cost Analysis view is configured for monthly service analysis.
- The anomaly rule Daily anomaly by resource group is Active and configured for email notification.
- The Cost alerts landing page may show “No Alerts to display” when no alert has actually triggered; this does not invalidate the configured rule.
- The current Cost Analysis screenshot shows no cost reported for September 2026, so actual top-five cost-driver values are not claimed until Azure reports genuine cost data.

15. Pay-As-You-Go Services and Project Usage

If the project is moved from a Free/Student demonstration to a Pay-As-You-Go Azure subscription, charges depend on actual resource usage and the selected configuration. The academic implementation should remain small and should avoid unnecessary paid resources.

Azure Service / Resource	Usage in This Project	PAYG Usage / Meter	Benefit
Storage Account	Stores the demonstration resource and related data.	Stored capacity and operations as a real billable resource for cost monitoring.	Provides a real billable resource for cost monitoring.
Cost Management	Analyzes subscription spending and budget.	Depends on Azure billing/account configuration.	Centralized cost visibility.
Budget Alerts	Sends threshold notifications.	Monitoring is tied to subscription cost.	Early warning of budget consumption.
Anomaly Alert	Checks configured anomaly view and sends notifications.	Depends on configuration of cost data available.	Identifies unusual spending patterns.

16. Final Service Selection for the Academic Project

Service	Core Project?	Account / Usage Approach
Azure Cost Management	Yes	Use with Azure for Students subscription.
Budgets	Yes	MonthlyCostBudget at ₹10 for the current demonstration.
Cost Analysis	Yes	Use Monthly Cost by Service; report actual values only when available.
Cost Alerts / Anomaly	Yes	Use active anomaly rule and threshold email notifications.
Storage Account	Yes	Use small StorageV2 resource in East Asia.
Tags	Yes	Apply Department, Project, and Owner tags.

17. Final Validation Checklist

- Resource group RG-CostManagement exists and contains the project resource.
- Storage account costalertstorage001 is available and correctly tagged.
- MonthlyCostBudget is configured at ₹10 per month.
- 80% (₹8) and 100% (₹10) actual-cost thresholds are configured.
- Cost Analysis is configured for monthly service-level review.
- Daily anomaly by resource group is shown as Active.
- Email recipients are configured for budget/anomaly notifications.
- Screenshots are collected as evidence for the academic project.
- No fabricated cost or top-five driver values are used when Azure reports no current cost data.